Review of Chapter 2.

§2.1.

1. General population Model.

$$\frac{dP}{dt} = (\beta - \delta) P.$$

where $\beta = \beta(t)$, $\delta = \delta(t)$, $P = P(t)$.

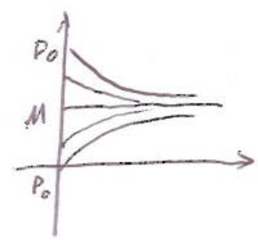
2. <u>Logistic equation</u> (Bounded Population)

$$\frac{dP}{dt} = aP - bP^2.$$

where a, b are both positive constants

Initial value problem

$$\begin{cases} \frac{dP}{dt} = kP(M-P) \\ P(0) = P_0 \end{cases} \rightarrow a = kM, \ b = k.$$

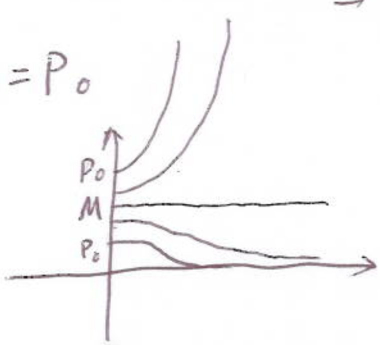


$$\Rightarrow P(t) = \frac{MP_0}{P_0 + (M-P_0)e^{-kMt}}$$

$$\lim_{t \to +\infty} P(t) = \frac{MP_0}{P_0 + 0} = M.$$ — limiting population.

3. Doomsday vs. Extinction.

$$\begin{cases} \frac{dP}{dt} = kP(P-M) \\ P(0) = P_0 \end{cases} \Rightarrow P(t) = \frac{MP_0}{P_0 + (M-P_0)e^{kMt}}.$$



i) $P_0 < M$

$$\lim_{t \to +\infty} P(t) = 0$$

ii) $P_0 > M$

$$\lim_{t \to +\infty} P(t) = +\infty$$

M – threshhold population.

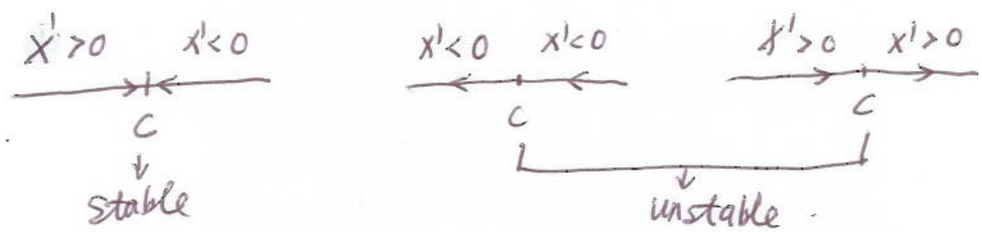
§2.2.

Autonomous first-order differential equation

$$\frac{dx}{dt} = f(x)$$

Defn: The solution of $f(x)=0$ are called <u>critical points</u>, If $x=c$ is a critical point, then $x(t)\equiv c$ is called an equilibrium solution of $\frac{dx}{dt}=f(x)$

Defn: A critical point $x=c$ of $\frac{dx}{dt}=f(x)$ is said to be <u>stable</u> provided that if the initial value x_0 is sufficiently close to c, the $x(t)$ remains close to c for all $t>0$.



§2.3. Motion under gravitational force and resistence force.

$$m\frac{dv}{dt} = \underbrace{F_G}_{\text{Gravitational force}} + \underbrace{F_R}_{\text{resistence force}} \qquad F_R = -kv^p$$

where $1\le p \le 2$, k a constant.

$$\Rightarrow \ m\frac{dv}{dt} = -mg - kv^p$$

Case I: $\overset{p=1}{\left\{\begin{array}{l} m\frac{dv}{dt} = -kv - mg \\ v(0)=v_0 \end{array}\right.}$ $\Rightarrow v(t) = (v_0 + \frac{g}{\rho})e^{-\rho t} - \frac{g}{\rho}$

where $\rho = \frac{k}{m}$.

$\lim_{t\to+\infty} v(t) = -\frac{g}{\rho} \longrightarrow$ terminal speed $|v_\tau| = \frac{g}{\rho} = \frac{mg}{k}$.

Case II: <u>$p=2$.</u>

upperward motion: $\frac{dv}{dt} = -g - \rho v^2$ (Since $v > 0$)

$$v(t) = \sqrt{\frac{g}{\rho}} \tan\left(\sqrt{\frac{\rho}{g}}\, v_0 - \sqrt{\rho g}\, t\right)$$

downward motion: $\frac{dv}{dt} = -g + \rho v^2$ (Since $v < 0$)

$$v(t) = \sqrt{\frac{g}{\rho}} \tanh\left(c - t\sqrt{\rho g}\right) \quad \text{where } c = \tanh^{-1}\left(v_0\sqrt{\frac{\rho}{g}}\right).$$

§2.4.

Consider

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

$\boxed{\text{Euler's Algorithm}}$

<u>step size</u>: h.

<u>Iterative formula</u>: $\begin{cases} x_{n+1} = x_n + h \\ y_{n+1} = y_n + h \cdot f(x_n, y_n) \end{cases}$ $(n = 0, 1, 2, \cdots k)$.

$$\Rightarrow \quad y_n \simeq y(x_n)$$

In fact

$$|y_n - y(x_n)| \le Ch \quad \text{for each } n = 1, 2, 3, \cdots, k.$$

Review of Chapter 3.

§3.1.

1. General form of second-order linear differential equation.

$$A(x)\, y'' + B(x)\, y' + C(x)\, y = F(x).$$

or

$$y''(x) + P(x)\, y' + q(x)\, y = f(x) \quad (\text{if } A(x) \neq 0)$$

i) $f(x) \equiv 0 \rightarrow$ homogeneous equation

ii) $f(x) \not\equiv 0 \rightarrow$ nonhomogeneous equation.

2. Let f, g be two functions defined on I.

<u>Wronskian</u> $W(f,g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = fg' - f'g$

Let y_1, y_2 be two solutions of $y'' + P(x)\, y' + q(x)\, y = 0$

i) y_1, y_2 linearly dependent $\Longleftrightarrow W(y_1, y_2) \equiv 0$ on I.

ii) y_1, y_2 linearly independent $\Longleftrightarrow W(y_1, y_2) \neq 0$ on every point of I.

3. General solution of homogeneous equation

$$y'' + P(x)\, y' + q(x)\, y = 0$$

is given by

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

where y_1, y_2 are any two linear independent solutions of

$y'' + P(x)\, y' + q(x)\, y = 0$

4. The case of constant coefficients

$$ay'' + by' + cy = 0$$

where a, b, c are constants.

<u>Characteristic equation:</u> $ar^2 + br + c = 0$

Let r_1, r_2 be two roots of the characteristic equation.

i) $r_1 \neq r_2$, real, then

$$y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

ii) $r_1 = r_2 = r$, real, then

$$y(x) = (C_1 + C_2 x) e^{rx}$$

iii) r_1, r_2 are complex conjugate pairs : $r_1 = a+bi$, $r_2 = a-bi$, then

$$y(x) = e^{ax}(C_1 \cos bx + C_2 \sin bx).$$

§3.2.

1. General nth-order Linear differential equation.

$$P_0(x) y^{(n)} + P_1(x) y^{(n-1)} + \cdots + P_{n-1}(x) y' + P_n(x) y = F(x)$$

i) $f(x) \equiv 0 \rightarrow$ homogeneous equation

ii) $f(x) \not\equiv 0 \rightarrow$ nonhomogeneous equation.

$$\longrightarrow y^{(n)} + P_1(x) y^{(n-1)} + \cdots + P_{n-1}(x) y' + P_n(x) y = f(x)$$

Initial Value problem

$$\begin{cases} y^{(n)} + P_1(x) y^{(n-1)} + \cdots + P_{n-1}(x) y' + P_n(x) y = f(x) \\ y(a) = b_0,\ y'(a) = b_1,\ \cdots,\ y^{(n-1)}(a) = b_{n-1} \end{cases} \Rightarrow \text{unique solution.}$$

2. Wronskian

$$W(f_1, f_2, \cdots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & & & \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix} \leftarrow \text{determinant of an } n\times n \text{ matrix.}$$

Let $y_1, y_2, \cdots, y_n$ be n solutions of the homogenous equation, then

i) $y_1, y_2, \cdots, y_n$ linearly independent $\Leftrightarrow W(y_1, \cdots, y_n) \neq 0$ on I.

ii) $y_1, y_2, \cdots, y_n$ linearly dependent $\Leftrightarrow W(y_1, \cdots, y_n) \equiv 0$ on I.

3. General solution of $y^{(n)} + P_1(x)y^{(n-1)} + \cdots + P_{n-1}(x)y' + P_n y = 0$ is given by

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$$

where $y_1(x), y_2(x), \cdots, y_n(x)$ are n linearly independent solution of the homogeneous equation.

§ 3.3.

Consider

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0 \rightarrow \text{homogeneous equation}$$

where $\{a_i\}_{i=0}^{n}$ are real constants.

Characteristic equation:

$$a_n r^n + a_n r^{n-1} + \cdots + a_2 r^2 + a_1 r + a_0 = 0.$$

roots: $r_1, r_2, \cdots, r_n$

i) distinct roots

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} + \cdots + c_n e^{r_n x}$$

ii) Repeated roots.

root: $\underline{r}$ with multiplicity $\underline{k}$, then the part of general solution corresponding to r is of the form

$$(c_1 + c_2 x + c_3 x^2 + \cdots + c_k x^{k-1}) e^{rx}$$

iii) Complex root

unrepeated pair of complex conjugate roots $r = a \pm bi$ the the part of the general solution corresponding to $a \pm bi$ is of the form

$$e^{ax}(c_1 \cos bx + c_2 \sin bx).$$

§ 3.4.

1. A mass-spring-dashpot system

$mx'' = F_S + F_R + \underbrace{F_E}_{}$ → external force.

F_S ↓ Restorative force

F_R ↳ force by dashpot

$F_S = -kx$ (Hooke's law).

$F_R = -cv = -cx'$

$F_E = f(t)$

$\Rightarrow \quad mx'' = -kx - cx' + f(t)$

$mx'' + cx' + kx = f(t)$

$c = 0 \rightarrow$ undamped motion $\quad c > 0 \rightarrow$ damped motion.

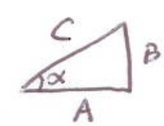
$f(t) \equiv 0 \rightarrow$ free motion $\quad f(t) \not\equiv 0 \rightarrow$ forced motion.

3. Free undamped motion.

$$mx'' + kx = 0$$

$$\Rightarrow x(t) = A\cos w_0 t + B\sin w_0 t \quad \text{where } w_0 = \sqrt{\frac{k}{m}}$$

$$= C\cos(w_0 t - \alpha) \quad \text{where}$$



$C = \sqrt{A^2 + B^2}$

$\cos\alpha = \frac{A}{C}, \ \sin\alpha = \frac{B}{C}$

$\tan\alpha = \frac{B}{A}$.

$C \rightarrow$ Amplitude.

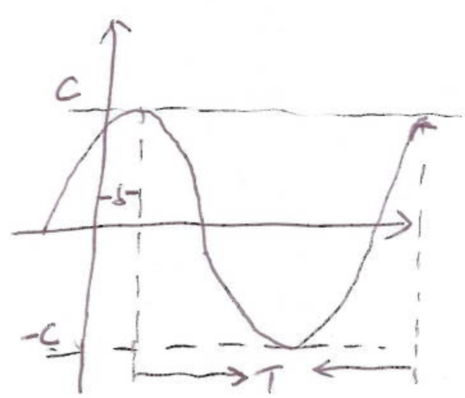
$w_0 \rightarrow$ Circular frequency.

$\alpha \rightarrow$ Phase angle.

Period : $T = \frac{2\pi}{w_0}$

frequency : $\nu = \frac{1}{T} = \frac{w_0}{2\pi}$

time lag : $\delta = \frac{\alpha}{w_0}$



§3.5.

1. <u>Nonhomogeneous equation</u>

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = f(x) \quad (Ly = f)$$

$$y = y_c + y_p$$

y_c - Complementary solution or say general solution of <u>the associated homogeneous equation</u> $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0 \quad (Ly = 0)$

y_p - a particular solution for the nonhomogeneous equation.

2. Method of Underdetermined Coefficient.

$f(x)$ is a linear combination of finite products of functions of the following three types.

1. A polynomial in X.
2. An exponential function e^{rx}
3. $\cos kx$ or $\sin kx$.

i). Let f be a sum of terms each of the form.

$$P_m(x) e^{rx} \cos kx \quad \text{or} \quad P_m(x) e^{rx} \sin kx. \qquad (*)$$

$\hookrightarrow$ <u>a polynomial in x of degree m.</u>

then take as the trial solution

$$y_p(x) = x^s \left[(A_0 + A_1 x + A_2 x^2 + \cdots + A_m x^m) e^{rx} \cos kx + (B_0 + B_1 x + B_2 x^2 + \cdots + B_m x^m) e^{rx} \sin kx \right]$$

where s is the smallest nonnegative integer such that no term in y_p duplicates a term in the complementary solution y_c.

Finally, determine the coefficients by substituting y_p into the nonhomogeneous equation.

ii) It is common to have.

$$f(x) = \underline{f_1(x)} + \underline{f_2(x)}.$$

have the form of ⊛.

then $y_p(x) = y_p^1(x) + y_p^2(x)$

Typical examples

$f(x)$	y_p
$P_m = b_0 + b_1 x + b_2 x^2 + \cdots + b_m x^m$	$x^s(A_0 + A_1 x + A_2 x^2 + \cdots + A_m x^m)$
$a\cos kx + b\sin kx$	$x^s(A\cos kx + B\sin kx)$
$e^{rx}(a\cos kx + b\sin kx)$	$x^s e^{rx}(A\cos kx + B\sin kx)$
$P_m(x)e^{rx}$	$x^s(A_0 + A_1 x + A_2 x^2 + \cdots + A_m x^m)e^{rx}$
$P_m(x)(a\cos kx + b\sin kx)$	$x^s[(A_0 + A_1 x + \cdots + A_m x^m)\cos kx + (B_0 + B_1 x + \cdots + B_m x^m)\sin kx]$